

Design Verification Report

RISC-V Single-Cycle CPU

Top-Level Integration Testbench

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DUT: sc_cpu_top_level | Stimulus: program.mem (30 instructions)

Simulator: Icarus Verilog (iverilog -g2012)

Date: 2026-05-31 | Status: PASS

VERIFICATION RESULT: PASS (0 errors)

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1. Executive Summary

This report documents the functional verification of a single-cycle RISC-V CPU at the top level (module `sc_cpu_top_level`). The design under test (DUT) was exercised with a 30-instruction program and checked against two independent reference oracles. All checks passed: the DUT register file, data memory, and program counter matched the golden reference at every committed instruction, and the hand-derived end-of-program state matched as well.

Result: **PASS, 0 errors**. 28 instructions executed before clean halt; the design terminated without watchdog intervention.

This program is a strong functional test of the supported instruction subset: every instruction type is exercised, and critically, all three conditional branches (`beq`, `bne`, `blt`) are tested in both the taken and not-taken directions. A small number of coverage gaps remain and are listed in Section 6.

2. Design Under Test

The DUT is a non-pipelined, single-cycle RV32I-subset processor using a Harvard architecture (separate instruction and data memories). Each instruction completes in one clock cycle. The program counter is reset asynchronously and advances on the rising clock edge.

2.1 Supported Instruction Subset

Category	Instructions	Notes
R-type	<code>add</code> , <code>sub</code> , <code>and</code> , <code>or</code> , <code>slt</code>	funct7 bit 5 selects <code>sub</code> vs. <code>add</code>
I-type	<code>addi</code> , <code>andi</code> , <code>ori</code> , <code>slti</code>	Immediate arithmetic; <code>slti</code> decodable
Load / Store	<code>lw</code> , <code>sw</code>	Word-aligned, computed base + offset
Branch	<code>beq</code> , <code>bne</code> , <code>blt</code>	One-hot branch control; PC-relative

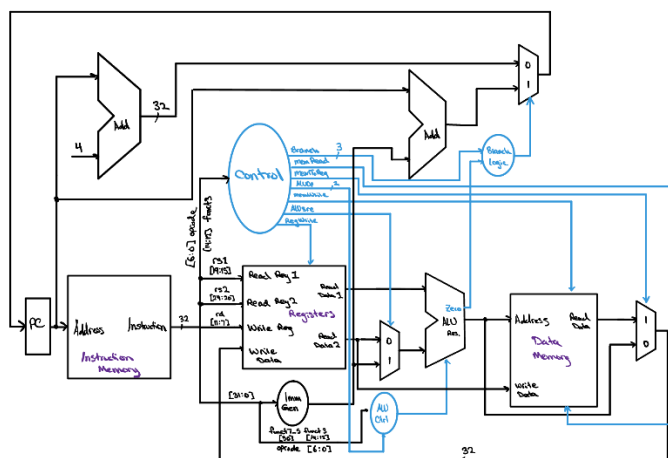


Figure 1: System Architecture

2.2 Module Hierarchy

sc_cpu_top_level	
├─ sc_cpu_control	Main control unit (combinational, opcode-decoded)
├─ alu_control	ALU control unit (funct3/funct7 decode)
├─ sc_cpu_datapath	
│ └─ instruct_mem	Read-only instruction memory (loaded from program.mem)
│ └─ imm_gen	Immediate generator (I, S, SB, U, UJ formats)
│ └─ reg_file	32x32-bit register file (2 read ports, 1 write port)
│ └─ mux_2x1	ALU source mux; write-back mux; branch/increment mux (x3)
│ └─ alu_full	32-bit ALU (ripple-carry, parameterized)
│ └─ alu_slice	1-bit ALU slice (bits [30:0])
│ └─ mux_2x1	Output select (ALU result vs. pass-through)
│ └─ mux_4x1	Operation select (add/sub, and, or, slt)
│ └─ full_adder	1-bit full adder
│ └─ alu_msb	MSB ALU slice (overflow detection, setless)
│ └─ mux_2x1	Output select (ALU result vs. pass-through)
│ └─ mux_4x1	Operation select (add/sub, and, or, slt)
│ └─ full_adder	1-bit full adder (MSB, feeds overflow logic)
├─ data_mem	Clocked data memory (word-aligned, read/write enable)
└─ ripple_carry_adder	PC + 4 adder; PC + immediate adder (x2)

3. Verification Methodology and Testbench

The testbench (`tb_sc_cpu_top_level`) is a self-checking, top-level integration testbench targeting Icarus Verilog (`iverilog -g2012`). Verification uses two independent oracles so that a bug in either the design or the checking code is unlikely to escape undetected.

3.1 Key Parameters

Parameter	Value / Purpose
CLK_PERIOD	10 ns — drives the always block that toggles clk
NUM_INSTR	30 — halt boundary; simulation stops when <code>ref_pc</code> word index ≥ 30
DMEM_WORDS	256 — number of 32-bit data-memory words checked each step
IMEM_WORDS	256 — reference instruction-memory size (mirrored from <code>program.mem</code>)
MAX_STEPS	1000 — runaway watchdog limit

3.2 Lockstep Golden Model

A behavioral RV32I-subset simulator (task `ref_step`) decodes and executes one RV32I instruction per call, maintaining its own `ref_regs[0:31]`, `ref_dmem[0:255]`, and `ref_pc`. It runs one instruction ahead of each DUT clock edge, encoding the correct architectural semantics. After every posedge `clk` the task `check_state` compares the DUT PC (`dut.datapath.pc`), all 32 register-file entries (`dut.datapath.registers.RF[k]`), and all 256 data-memory words (`dut.datapath.data_memory.memory[k]`) against the reference. Any mismatch is logged immediately with step number and field name.

3.3 Final-State Oracle

The tasks `load_expected` and `check_final` encode a hand-derived, independently computed table of the expected end-of-program register and memory state, checked once after halt. Because it

is produced by a completely different method from the lockstep model (manual trace), agreement among DUT, lockstep model, and this table gives high confidence that a latent bug in the lockstep model itself has not masked a DUT defect.

3.4 Supported Instruction Subset

The golden model (task `ref_step`) decodes and executes the following RV32I instructions:

Opcode group	Instructions	Notes
OPC_RTYPE (0110011)	add, sub, and, or, slt	funct7[5] selects sub; signed compare for slt
OPC_ITYPE (0010011)	addi, andi, ori, slti	12-bit sign-extended immediate
OPC_LOAD (0000011)	lw	Byte address = $rs1 + imm_i$; word-aligned
OPC_STORE (0100011)	sw	Byte address = $rs1 + imm_s$; word-aligned
OPC_BRANCH (1100011)	beq, bne, blt	PC-relative; signed compare for blt

3.5 Halt Condition and Watchdog

The main execution loop terminates when `ref_pc[31:2] >= NUM_INSTR (30)`. Because the DUT uses the same program image and identical control flow, it reaches the same halt condition on the same clock edge. A global watchdog (initial block with absolute delay) triggers `$finish` after `CLK_PERIOD * (MAX_STEPS + 50)` nanoseconds if the loop has not already exited, preventing the simulator from hanging on a runaway design.

3.6 Waveform Dump

The testbench calls `$dumpfile("waveforms/dump.vcd")` and `$dumpvars(0, tb_sc_cpu_top_level)` in an initial block, producing a Value Change Dump covering the full hierarchy. The VCD is viewable in EPWave (EDA Playground) or any compatible waveform viewer such as GTKWave.

3.7 Per-Instruction Trace and Disassembler

After each step the task `trace_print` emits a formatted line containing the step counter, PC (hex), instruction word (hex), a pseudocode disassembly produced by the task `disasm`, and a plain-English description of the architectural effect (register write, store, branch taken/not taken, or no state change). The `disasm` task covers all opcodes in the supported subset and falls back to `.word 0xxxxxxxxx` for any unrecognized encoding.

3.8 Hierarchical References

The testbench accesses DUT internal state directly via hierarchical dot-notation paths, which must match the actual module and signal names in the RTL:

<code>dut.datapath.pc</code>	– program counter
<code>dut.datapath.registers.RF[k]</code>	– register file array ($k = 0..31$)
<code>dut.datapath.data_memory.memory[k]</code>	– data memory array ($k = 0..255$)

If the RTL uses different instance or signal names, these three paths must be updated accordingly for the testbench to compile and link.

3.9 Verification Task Reference

Task	Role
disasm	Decode one 32-bit instruction word into pseudocode; covers all supported opcodes, falls back to .word for unknowns
ref_step	Decode and execute one instruction in the behavioral model; update ref_regs, ref_dmem, ref_pc; set trace variables
check_state	Compare DUT PC, all 32 registers, and all 256 data-memory words against the reference; print any mismatch
trace_print	Emit a human-readable per-instruction trace line: PC, hex encoding, pseudocode disassembly, effect
dump_final_state	Print the complete register file and all non-zero data-memory words at end of simulation
load_expected	Populate the hand-derived final-state oracle (exp_regs, exp_dmem)
check_final	Compare DUT final state against the oracle; report PASS or error count

4. Program Under Test

The 30-instruction program (`program.mem`) is loaded into both the DUT instruction memory and the reference model at simulation start via `$readmemh`. The execution trace below shows every committed instruction, the disassembled pseudocode, and the architectural effect. For detailed waveshape, see Appendix A.

4.1 Execution Trace

Step	PC	Encoding	Pseudocode	Effect
1	0x00000000	0x00100093	<code>addi x1, x0, 1</code>	$x1 \leftarrow 0x00000001$ (1)
2	0x00000004	0x00500113	<code>addi x2, x0, 5</code>	$x2 \leftarrow 0x00000005$ (5)
3	0x00000008	0x002081B3	<code>add x3, x1, x2</code>	$x3 \leftarrow 0x00000006$ (6)
4	0x0000000C	0x40118233	<code>sub x4, x3, x1</code>	$x4 \leftarrow 0x00000005$ (5)
5	0x00000010	0xFFE20293	<code>addi x5, x4, -2</code>	$x5 \leftarrow 0x00000003$ (3)
6	0x00000014	0x0042E333	<code>or x6, x5, x4</code>	$x6 \leftarrow 0x00000007$ (7)
7	0x00000018	0x005273B3	<code>and x7, x4, x5</code>	$x7 \leftarrow 0x00000001$ (1)
8	0x0000001C	0x00522433	<code>slt x8, x4, x5</code>	$x8 \leftarrow 0x00000000$ (0) — $5 < 3$ false
9	0x00000020	0x0042A4B3	<code>slt x9, x5, x4</code>	$x9 \leftarrow 0x00000001$ (1) — $3 < 5$ true
10	0x00000024	0x00302223	<code>sw x3, 4(x0)</code>	$\text{dmem}[1] \leftarrow 0x00000006$
11	0x00000028	0x00800513	<code>addi x10, x0, 8</code>	$x10 \leftarrow 0x00000008$ (8)
12	0x0000002C	0x00652223	<code>sw x6, 4(x10)</code>	$\text{dmem}[3] \leftarrow 0x00000007$
13	0x00000030	0x407105B3	<code>sub x11, x2, x7</code>	$x11 \leftarrow 0x00000004$ (4)
14	0x00000034	0xFFC52603	<code>lw x12, -4(x10)</code>	$x12 \leftarrow \text{dmem}[1] = 0x00000006$ (6)

Step	PC	Encoding	Pseudocode	Effect
15	0x00000038	0x00410663	beq x2, x4, +12	beq: x2=5 == x4=5 → TAKEN → 0x00000044
—	0x0000003C	0x003002B3	add x5, x0, x3	dead code — skipped by branch at step 15
—	0x00000040	0x00F17693	andi x13, x2, 15	dead code — skipped by branch at step 15
16	0x00000044	0x00A02823	sw x10, 16(x0)	dmem[4] ← 0x00000008
17	0x00000048	0x00F17713	andi x14, x2, 15	x14 ← 0x00000005 (5)
18	0x0000004C	0x00411A63	bne x2, x4, +20	bne: x2=5 != x4=5? NOT taken
19	0x00000050	0x00311463	bne x2, x3, +8	bne: x2=5 != x3=6 → TAKEN → 0x00000058
—	0x00000054	0x407105B3	sub x11, x2, x7	dead code — skipped by branch at step 19
20	0x00000058	0x00610233	add x4, x2, x6	x4 ← 0x0000000C (12)
21	0x0000005C	0xFE1048E3	blt x0, x1, -16	blt: x0=0 < x1=1 → TAKEN → 0x0000004C
22	0x0000004C	0x00411A63	bne x2, x4, +20	bne: x2=5 != x4=12 → TAKEN → 0x00000060
23	0x00000060	0x00326793	ori x15, x4, 3	x15 ← 0x0000000F (15)
24	0x00000064	0x00208663	beq x1, x2, +12	beq: x1=1 == x2=5? NOT taken
25	0x00000068	0x00114463	blt x2, x1, +8	blt: x2=5 < x1=1? NOT taken
26	0x0000006C	0x00902A23	sw x9, 20(x0)	dmem[5] ← 0x00000001
27	0x00000070	0x00928013	addi x0, x5, 9	x0 write discarded (result = 0x0000000C)
28	0x00000074	0x40238833	sub x16, x7, x2	x16 ← 0xFFFFFFF C (-4)

Instructions at PCs 0x3C, 0x40, and 0x54 are dead code: the branches at steps 15 and 19 cause execution to skip them entirely in this deterministic program.

5. Verification Results

5.1 Lockstep Check

The lockstep check fires after every clock edge and compares the DUT PC, all 32 registers, and all 256 data-memory words against the golden model. Zero mismatches were recorded across all 28 executed steps.

5.2 Final-State Oracle

The hand-derived oracle was compared against the DUT final state after halt. Zero mismatches were recorded. The complete expected and observed state is shown below.

5.3 Final Register File

Register	Expected (hex)	Observed (hex)	Comment
x0	0x00000000	0x00000000	Hardwired zero — write at step 27 discarded
x1	0x00000001	0x00000001	addi x1, x0, 1
x2	0x00000005	0x00000005	addi x2, x0, 5
x3	0x00000006	0x00000006	add x3, x1, x2
x4	0x0000000C	0x0000000C	add x4, x2, x6 (step 20)
x5	0x00000003	0x00000003	addi x5, x4, -2
x6	0x00000007	0x00000007	or x6, x5, x4
x7	0x00000001	0x00000001	and x7, x4, x5
x8	0x00000000	0x00000000	slt x8, x4, x5 — false; not in non-zero set
x9	0x00000001	0x00000001	slt x9, x5, x4 — true
x10	0x00000008	0x00000008	addi x10, x0, 8
x11	0x00000004	0x00000004	sub x11, x2, x7
x12	0x00000006	0x00000006	lw x12, -4(x10) — loads dmem[1] = 6
x13	0x00000000	0x00000000	Dead code — andi x13 skipped
x14	0x00000005	0x00000005	andi x14, x2, 15
x15	0x0000000F	0x0000000F	ori x15, x4, 3
x16	0xFFFFFFF0	0xFFFFFFF0	sub x16, x7, x2
x17–x31	0x00000000	0x00000000	Not written

5.4 Final Data Memory (non-zero words)

Word index	Byte address	Expected (hex)	Observed (hex)	Source instruction
1	0x04	0x00000006	0x00000006	sw x3, 4(x0)
3	0x0C	0x00000007	0x00000007	sw x6, 4(x10)
4	0x10	0x00000008	0x00000008	sw x10, 16(x0)
5	0x14	0x00000001	0x00000001	sw x9, 20(x0)

6. Functional Coverage

6.1 Coverage Matrix

Coverage Point	Status	Notes
R-type: add	COVERED	Steps 3, 20
R-type: sub	COVERED	Steps 4, 13, 28
R-type: and	COVERED	Step 7
R-type: or	COVERED	Step 6
R-type: slt (false result)	COVERED	Step 8
R-type: slt (true result)	COVERED	Step 9
I-type: addi (positive imm)	COVERED	Steps 1, 2, 11
I-type: addi (negative imm)	COVERED	Step 5 (-2)
I-type: addi to x0 (discard)	COVERED	Step 27: addi x0, x5, 9 — write suppressed
I-type: andi	COVERED	Step 17 (andi x14)
I-type: ori	COVERED	Step 23
I-type: slti	NOT COVERED	Decodable but not present in program
Load: lw (negative offset)	COVERED	Step 14: lw x12, -4(x10) → loads dmem[1]
Store: sw (positive offset)	COVERED	Steps 10, 12, 16, 26
Branch: beq taken	COVERED	Step 15: beq x2, x4 (both = 5)
Branch: beq not taken	COVERED	Step 24
Branch: bne taken	COVERED	Steps 19, 22
Branch: bne not taken	COVERED	Step 18: bne x2, x4 (both = 5)
Branch: blt taken	COVERED	Step 21: blt x0, x1 (0 < 1)
Branch: blt not taken	COVERED	Step 25: blt x2, x1 (5 < 1 false)
x0 write suppression	COVERED	Step 27 confirms hardwired-zero behaviour
Registers x17–x31 written	NOT COVERED	Only x0–x16 are touched
Data memory word index > 5	NOT COVERED	Only indices 1, 3, 4, 5 written

6.2 Dead-Code Lines

Three instructions are never executed: `add x5, x0, x3` at PC `0x3C`, `andi x13, x2, 15` at PC `0x40`, and `sub x11, x2, x7` at PC `0x54`. The first two are skipped by the taken branch at step 15; the third is skipped by the taken branch at step 19. All three are present in the program image and decoded by the reference disassembler but contribute no architectural effect.

7. Issues and Recommendations

7.1 Open Issues

#	Issue	Details
1	slti not exercised	The I-type ALU slice for slti (funct3=010) is present in both the RTL and the golden model but is never triggered by the current program. A one-instruction extension to the test program would close this gap.
2	Upper registers unexercised	Registers x17 through x31 are never written. A follow-on test that exercises the full register file would increase confidence in the register-file address decoding.
3	Data memory sparsely exercised	Only word indices 1, 3, 4, and 5 are written. A supplementary program that exercises boundary indices (0, 255) and adjacent words would improve confidence in the address decoder.

7.2 Recommendations

- Supplement program.mem with a short follow-on sequence that covers slti and at least one upper register (x17+).
- Consider adding a ``timescale` directive to each RTL module to eliminate the Icarus Verilog timing-precision warning that appears when mixed-timescale designs are compiled together.
- Add a fault-injection run: force one register to a wrong value mid-simulation and confirm that the lockstep check flags it immediately.
- Extend data-memory testing to include word index 0 and the upper boundary (index 255) to stress the address decoder.

8. Conclusion

The single-cycle RISC-V CPU (`sc_cpu_top_level`) passes all verification checks on the 30-instruction test program. The dual-oracle testbench reported zero errors from the lockstep comparison and zero errors from the independent hand-derived final-state check. The design correctly implements addition, subtraction, bitwise AND/OR, set-less-than, immediate arithmetic, load-word (negative offset), store-word, and all six branch cases (`beq/bne/blt`, both taken and not-taken). The x0 write-suppression path was also exercised. The remaining coverage gaps are minor and are addressed by the recommendations in Section 7.

Appendix A. Simulation Waveform

The waveform below shows the DUT signal activity captured during the 30-instruction simulation run. Signals are grouped by function: clock, reset, and pc (yellow), instruction decode (violet), control (green), data words (red), and ALU control signals (orange).

